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**SALAMLEARN: AN INTERACTIVE ARABIC LEARNING APP FOR
FILIPINO MUSLIM CHILDREN**

A Capstone Project Title Proposal
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CHAPTER I

1.1 INTRODUCTION

Worldwide, education shifts steadily due to progress in how people share and receive information. Instead of only textbooks and lectures, tools like tablets, apps, and online classrooms now appear more often inside learning spaces. These digital supports offer ways to involve students through responsive tasks, simple access, and formats that hold attention. For younger minds especially, such tech-backed settings tend to increase drive, involvement, and memory strength during lessons, better than standard lecture-only paths (Lampropoulos & Kinshuk, 2024). Because of this trend, schools look repeatedly toward new blends of digital resources within instruction, adjusting to what today's students expect and require.

From all areas where education meets digital tools, picking up languages stands out due to shifts toward smartphone-supported methods. Learners find themselves able to practice at their own speed using apps designed for interaction outside school walls. Evidence suggests mixing audiovisual material with game-like tasks supports better focus and skill growth across seeing, hearing, and doing modes of study (Figueiredo, 2023; Wiboolyasarín & Jinowat, 2024). When learning feels like play, internal drive rises - especially among younger users drawn to hands-on tasks paired with incentives (Li et al., 2024; Alsadoon et al., 2022).

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In the Philippines, Madrasah schools help young Muslims learn Arabic, practice faith-based values, and study religion. Starting early, students become familiar with Arabic letters, correct sounds, prayer routines, together with core beliefs through structured lessons. Yet most classrooms still depend heavily on manual learning, verbal repetition, instructor-led examples, alongside paper-based resources. Although such techniques sustained religious teaching across decades, they sometimes fail to match how today’s youth engage with information. Digital exposure grows steadily among children, bringing new expectations about interaction and pacing in learning experiences. Few tools exist locally that combine technology with beginner-level Arabic reading skills tailored for Muslim pupils in this region.

It appears newer research shows digital tools may strengthen how involved students feel during lessons, especially if used alongside standard teaching methods (Lampropoulos & Kinshuk, 2024; Liu et al., 2023). When it comes to studying Arabic, techniques involving interaction or game-like elements tend to increase involvement while creating more favorable conditions for learning (Almelhes, 2024). What emerges is a picture where technology, though not central, could assist both school settings and independent study outside classrooms.

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	Responding to such educational demands, SalamLearn emerges, a digital tool crafted for young Filipino Muslims beginning Arabic. This application delivers customized interaction through a mobile interface shaped by local context. Instead of generic methods, it builds skills via guided writing practice for Arabic script. Pronunciation receives structured support, helping users form sounds with growing accuracy. Embedded games link religious knowledge with active participation, avoiding passive reception. Prayer movements appear through dynamic visual instruction, offering clarity without reliance on internet access. Achievement tracking unfolds quietly behind milestones, encouraging consistent effort. Functionality remains intact even when connectivity fails, classroom mode operates independently. By blending pedagogy with cultural resonance, the design nurtures early-stage literacy in sacred language. Engagement rises naturally where relevance meets accessibility. 1.2 PROJECT CONTEXT Despite their role in shaping early religious understanding, Madrasah schools typically rely on repetition teaching techniques alongside paper-based resources. Instruction usually begins with exposure to Arabic script, correct articulation, worship routines, and core beliefs, guided by teachers through repetition and oral practice. These centers provide spaces where young Filipino Muslims develop initial familiarity with sacred texts and spiritual principles. Yet such time-honored strategies, while significant, tend to prioritize memorization over exploration. New findings suggest digital tools designed for interaction may	
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encourage deeper involvement by adapting to how individuals learn (Li et al., 2024; Alsadoon et al., 2022). The engagement gap and pedagogical problems identified are supported by initial observations from local Madrasah teachers and preliminary needs assessments, which will be further solidified during formal data collection.

Children now face greater contact with screens and digital platforms, opening paths to blend standard teaching methods with tech-based tools. Research into phone-supported education shows apps using sight, sound, and movement help build reading skills, language growth, because they engage multiple senses at once (Figueiredo, 2023; Wiboolyasarín & Jinowat, 2024). In similar fashion, game-like features, badges, progress markers, tasks requiring response, tend to boost involvement during study sessions among early-age users (Jogo et al., 2022; Li et al., 2024). Still, even as schools adopt more software for classrooms, few programs exist in regional formats aimed at Arabic reading ability or foundational religious knowledge for young Muslims in the Philippines. Most available options serve global markets instead, failing to reflect specific lesson goals, community norms, daily routines common within native Madrasah settings.

A further difficulty observed in certain Madrasah settings involves restricted access to personal digital tools for particular students. Classroom policies restricting phone use during lessons add complexity. When learning depends on private gadgets, such constraints risk reducing impact. Oversight of student development beyond standard tests remains difficult for caregivers and teachers alike. Evidence indicates systems using activity logs and responsive

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feedback loops help shape better academic decisions (Liu et al., 2023; Lampropoulos & Kinshuk, 2024).

To address these problems, SalamLearn is proposed as a mobile-based educational platform specifically designed for Filipino Muslim children. SalamLearn integrates an Interactive Character Module for Arabic tracing exercises, an Arabic Pronunciation Support Module for audio-assisted learning, an Islamic Rules Gamification Module for teaching Halal and Haram concepts, an Interactive Salah Guide for prayer procedure visualization, and a Progress Rewards System that encourages continuous engagement. The application also incorporates an Offline-First Madrasah Classroom Mode that enables educators to conduct interactive learning sessions through televisions, projectors, or shared devices without requiring internet connectivity or individual student smartphones. In addition, SalamLearn provides collaborative progress-monitoring capabilities that allow parents to monitor their children's learning at home while enabling verified teachers to organize class sections, monitor learner progress, and assign existing SalamLearn learning activities. Through these features, SalamLearn supports both independent home learning and school-integrated instruction while remaining engaging, accessible, culturally relevant, and aligned with the DepEd ALIVE curriculum.

1.3 PROJECT DESCRIPTION

SalamLearn is an offline-first, interactive mobile application designed to augment the traditional pedagogical methods of the Philippine Madrasah system. The system architecture is specifically bound to align with the Department of Education's (DepEd) Refined Elementary Madrasah Curriculum (ALIVE) for Grade 1 learners (ages 5 to 10). To address localized constraints, such as intermittent internet connectivity, shared device usage, and strict adherence to the Data Privacy Act of 2012 (RA 10173), the application utilizes a local caching framework (Hive) alongside a unified codebase (Flutter).

The application is structured into four integrated user environments: the Student Hub for independent learning, the Parent Dashboard for home-based progress monitoring, the Teacher Learning Module for classroom organization and learner monitoring, and the Offline Classroom Mode for one-to-many institutional teaching. Together, these environments support both independent home learning and school-integrated instruction while sharing the same educational content and learner progress records. These environments are driven by nine comprehensive system modules:

1. Onboarding and System Initialization Module

This operational module handles device readiness and initial user configuration.

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- **System Readiness Checker:** A background protocol executed during the splash screen that verifies local database integrity and ensures the device has the minimum required storage capacity (500MB) to cache heavy audio and visual assets.
- **Localization Setup:** A user interface toggle that establishes the primary instructional language (English or Filipino) for the application's non-Arabic text.

2. Authentication and Profile Management Module

Acting as the primary security gateway, this module ensures compliance with RA 10173 regarding the collection of minor-generated telemetry data.

- **Parental Gate and Consent:** A secure registration/login interface (Firebase) with local offline caching, requiring explicit adult consent for local data storage and processing.
- **Profile and PIN Establishment:** Allows the parent to generate a child profile and configure a 4-digit numeric PIN, which encrypts and restricts access to all administrative and institutional areas of the application.

3. The Student Hub (Child Interface)

The primary graphical user interface (GUI) designed strictly for early childhood Human-Computer Interaction (HCI) standards. It features high-contrast visual elements, minimal text, and large icon-based touch targets to reduce

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	4. Core Educational Modules (Grade 1 ALIVE Curriculum) This module translates the DepEd ALIVE syllabus into gamified, tactile, and audio-visual interactive components: ● Arabic Tactile Recognition (Language): An algorithmic path-tracing interface that validates right-to-left single-finger strokes for the 28 base Arabic letters, numerals, and basic shapes. ● Pronunciation & Vocabulary Builder (Language): A tap-and-listen digital flashcard system triggering offline, high-quality native audio for basic Islamic greetings and Grade 1 vocabulary themes (e.g., body parts, colors). ● Qur'an & Hadith Audio Hub: An interactive auditory module for the guided memorization of Surah al-Fatihah and the seven specific Grade 1 Suwar, alongside contextual touch-targets for daily supplications (e.g., waking up, eating). ● Interactive Sirah & Aqidah Storybook: A structured, linear visual storyboard detailing the early life of Prophet Muhammad (Sirah) and identifying the fundamental attributes of Allah (Aqidah) without requiring heavy reading comprehension. ● Taharah & Fiqh Gamification: Logic-based sorting and sequencing games, including a drag-and-drop "5 Pillars" matching activity, a step-by-step Wudhu (Ablution) sequencing simulator, and a clean versus impure water sorting exercise.	
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5. Progress and Rewards System

A positive reinforcement subsystem that monitors module completion. It triggers real-time visual feedback (e.g., animations, chimes) during successful tactile interactions and populates a "Digital Backpack" with achievement badges tied to specific Grade 1 milestones (e.g., "Wudhu Master").

6. Parent Analytics Dashboard (PIN Protected)

A secure monitoring interface designed to facilitate parental involvement and active mediation at home.

- **Telemetry Visualization:** Processes raw SQLite/Hive gameplay data (tracing accuracy, sequencing errors, time-on-task) into intuitive visual graphs.
- **Mediation Prompts:** An automated system that generates actionable, curriculum-aligned suggestions for parents based on the child's identified weak points (e.g., suggesting specific physical practice for Wudhu steps).

7. Teacher Learning Module

A secure educator interface designed to support classroom integration while preserving SalamLearn's focus as an interactive learning platform rather than a Learning Management System

- **Classroom Section:** Allows verified teachers to create and manage multiple class sections (Grade 1 - Section A, Grade 1 - Section C),

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organize enrolled learners, and generate invitations codes for students and parents to join a class

- **Student Progress Monitoring:** Displays learner completion rates, tracing accuracy, pronunciation performance, lesson mastery, badges earned, and recent learning activity through summarized classroom analytics
- **Practice Assignment:** Enables teachers to assign existing SalamLearn educational modules to an entire class for home practice. Teachers cannot create custom lessons or upload external learning materials.

8. Offline-First Madrasah Classroom Mode (PIN Protected)

A specialized 1-to-many presentation interface designed for the Asatidz (teachers) to bypass the 1-to-1 device limitation in public Madrasah classrooms.

- **Presenter UI and Screen Casting:** Scales the application's GUI for external display via projectors or smart TVs, granting the teacher manual override controls over all modules.
- **Choral Reading & Flashcard Controller:** Allows the teacher to trigger specific audio files for whole-class Talqeen (repetition) without game progression locks.
- **"Hot Seat" Gamification:** Enables the teacher to project the Tactile Tracing or Wudhu Simulator modules onto the board while a single student volunteer solves the sequence on the host device.

9. Settings and System Maintenance (PIN Protected)

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	The administrative control center for device and data management. ● Media Mixing: Independent volume controls for background audio, UI sound effects, and pronunciation voices to accommodate varying sensory needs. ● Data Lifecycle Management: Features manual synchronization triggers for pushing local offline data to the cloud when Wi-Fi is detected, and a secure data-erasure protocol to satisfy the "Right to be Forgotten" under the Philippine Data Privacy Act.	
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1.4 OBJECTIVES OF THE STUDY

The primary objective is to Develop **“SalamLearn: An Interactive Arabic Learning App for Filipino Muslim Children”**. Specifically, the project aims to: Specifically, the project aims to:

9. Design a comprehensive interactive learning system incorporating the following core functionalities
 - a. **Arabic Tactile Recognition Module:** Design an interface for identifying foundational Arabic symbols and characters through interactive visual tracing.
 - b. **Rules Gamification Module:** Develop a categorization game to help children recognize and sort concepts such as Halal and Haram.
 - c. **Daily Habit Guide:** Implement a step-by-step visual walkthrough for recognizing daily religious duties and sequences.
 - d. **Offline Classroom Package:** Develop a teacher-assisted classroom mode that allows educators to deliver interactive Arabic and Islamic learning activities through projectors, televisions, or shared devices without requiring internet connectivity or individual student smartphones.
 - e. **Teacher Learning Module:** Develop a secure teacher interface that enables educators to organize class sections, monitor student learning progress, and assign existing SalamLearn learning

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activities while preserving the platform's focus on foundational Arabic and Islamic education.

- f. **Arabic Pronunciation Module:** Implement audio-assisted learning activities that enable children to hear and repeat correct Arabic pronunciations while interacting with lesson content.
- g. **Progress Rewards System:** Develop a motivation-based module that awards digital badges for consistent engagement and mastery of Arabic literacy topics.

10. Create the system using a modern and scalable technical architecture:

- a. **Front-end Development:** Build a responsive and child-friendly mobile interface using a cross-platform framework, specifically Flutter (Dart) to ensure accessibility on various devices.
- b. **Database:** Utilize local storage technologies such as Hive or SQLite to support offline functionality, particularly for the Offline Classroom Package. This allows lessons content, images, and audio assets to be accessed without internet connectivity during classroom instruction
- c. **Back-end Development:** Implement and utilize Firebase, which provides authentication, cloud-based database management, and real-time data synchronisation for tracking student progress and parent dashboard analytics.

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- d. **Asset Management:** Organize and integrate high-quality visual assets and localized Arabic content optimized for mobile performance.

11. Test and improve reliability and accuracy through rigorous testing phases:

- a. **Functional Testing:** Perform unit and integration testing on the gamification logic to ensure accuracy in feedback and rewards.
- b. **Usability Testing:** Conduct Alpha and Beta testing with children and parents to identify UI/UX bottlenecks and refine the navigation flow based on user feedback.
- c. **Compatibility Testing:** Evaluate the application's ability to function consistently across different Android devices, screen devices, Android operating system versions, and hardware specifications to ensure accessibility and stable performance for all intended users.

12. Evaluate system performance and impact using the ISO/IEC 25010 System and Software Quality Model:

- a. **Functional Suitability:** Assess whether the system provides all necessary interactive features to support foundational Arabic and religious literacy.
- b. **Usability:** Measure the learnability and operability of the interface for the target age group using standardized ease-of-use metrics.
- c. **Performance Efficiency:** Evaluate the application's responsiveness and stability during active learning sessions.

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- d. **Security:** Evaluate the system's capability to protect user account information, progress records, and profile data through secure authentication and controlled access mechanisms.
- e. **Reliability:** Assess the application's ability to maintain stable operation and preserve learning progress during regular use.

1.5 PROJECT SCOPE AND DELIMITATION

The primary scope of the SalamLearn platform is to deliver a localized, interactive learning experience tailored for Filipino Muslim children aged 5 to 10. The system's technical capabilities are concentrated in four main areas: an Arabic Literacy Module utilizing a visual-first interface for character identification through tactile path-tracing, a Jurisprudence Gamification Module employing sorting game logic to teach Halal and Haram concepts, a Progress Monitoring Component featuring real-time mastery tracking, and an Offline Classroom Package that supports teacher-led instruction through televisions, projectors, or shared classroom devices. Built on a modern cross-platform framework with a Firebase backend, the application focuses on providing an engaging, high-performance environment for foundational religious education. While Firebase is utilized for cloud-based synchronization of student progress, the primary operational data for learning materials and offline classroom activities is handled locally on the device through asset caching, ensuring that core functionality remains independent of internet connectivity. The system supports

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	two complementary modes of learning. Independent Learning Mode enables parents to create child profiles and facilitate Arabic and Islamic learning at home without requiring school enrollment. School-Integrated Mode enables verified teachers to organize learners into class sections, monitor student progress, assign existing SalamLearn practice activities, and utilize the Offline Classroom Mode during instruction. Both modes provide access to the same educational content while differing only in learner monitoring and classroom collaboration features. However, real-time student interaction during classroom sessions is limited to educator-managed activities and does not include individual student device synchronization during live lessons. To ensure feasibility within the academic timeframe, the project is bound by several limitations. The system is exclusively developed for the Android platform and does not include dedicated iOS or web-based versions. Content is strictly delimited to entry-level foundational literacy and basic religious rules, excluding advanced Quranic Tajweed, complex linguistic grammar, or deep theological debates. Furthermore, the application stores only localized progress data and profile settings, avoiding connections to external unverified religious databases. All religious content included within the application will be limited to materials reviewed and approved by designated Madrasah educators and content validators. The system does not generate religious rulings, interpretations, or theological opinions beyond the validated educational materials provided by the project team. While the Offline Classroom Package supports teacher-facilitated learning, classroom activities are managed entirely by the educator and do not	
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	include real-time synchronization of individual student responses. The classroom features are designed to enhance Arabic and Islamic instruction through learner progress monitoring and practice assignment. SalamLearn is not intended to function as a Learning Management System (LMS), Student Information System, or general classroom administration platform. Features such as attendance monitoring, messaging, document submission, curriculum authoring, grading, and file sharing are outside the scope of the proposed system. The system is subject to specific technical and operational constraints. Regarding device compatibility, the application requires a minimum of Android 8.0 (Oreo) and is optimized for tablet-sized screens (7 inches or larger) to facilitate effective tactile tracing, though it remains functional on smaller handsets. Due to extensive offline asset caching of high-quality audio and animations, the device must have at least 500MB of available internal storage capacity. Audio quality is fixed based on pre-validated, high-fidelity studio assets to ensure phonetic accuracy without real-time synthesis. Furthermore, the system is strictly dependent on validated content; all religious materials, guides, and moral classifications must be pre-approved by designated Madrasah educators and content validators prior to system integration to maintain theological integrity. Finally, SalamLearn is designed as an interactive Arabic and Islamic learning platform that complements both independent home learning and formal Madrasah instruction. The application is intended to strengthen collaboration among students, parents, and teachers through interactive learning experiences	
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	and progress monitoring, but it does not replace the instructional role of qualified Madrasah educators. 1.6 SIGNIFICANCE OF THE PROJECT The development of SalamLearn is anticipated to deliver significant value to the educational ecosystem of local Madrasah communities and the broader field of instructional technology. The following stakeholders are expected to benefit from this initiative: For the Students. Children will gain an interactive and rewarding learning environment through the Arabic Tactile Recognition, Arabic Pronunciation, and Rules Gamification modules. These features transform abstract concepts into engaging visual experiences, enabling students to achieve higher scores in character recognition assessments and improve retention of fundamental rules more effectively than through traditional memorization alone. For the Parents. The Parent Analytics Dashboard enables parents to actively monitor their children's learning progress, identify areas requiring additional practice, and reinforce Arabic and Islamic learning at home. For learners enrolled in participating Madrasah institutions, parents may also monitor progress alongside teacher-guided classroom activities, strengthening home-school collaboration.	
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	For the Academe and Madrasah Institutions. SalamLearn provides educators with an interactive learning platform that supports both classroom instruction and home-based reinforcement. Through the Teacher Learning Module, educators can organize class sections, monitor learner progress, assign existing SalamLearn practice activities, and utilize the Offline Classroom Mode during instruction. These features promote stronger collaboration between teachers and parents while preserving traditional Madrasah teaching practices. For Future Researchers. This study establishes a foundational framework for applying gamification and tactile-recognition logic in localized, faith-based educational tools. Future researchers may utilize the findings and technical architecture of SalamLearn as a baseline for extending interactive learning systems to other specialized cultural or linguistic disciplines.	
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1.7 PROJECT TIMELINE AND SCHEDULE OF ACTIVITIES

The project timeline for SalamLearn is structured into five different phases, spanning from the initial conceptualization to final institutional clearance.

Phase I: Methods of Research (January 2026 - May 2026) focuses on identifying the research gap and establishing the theoretical framework. This phase concludes with the Title Proposal defense, an approved capsule, and the acceptance of the selected Capstone Project adviser.

Phase II: Capstone Project 1 (June 2026 - July 2026) represents an intensive design phase. During this period, the focus shifts from defining objectives to implementation, with emphasis on system architecture, database schema design, and UI/UX wireframing for the SalamLearn application. The primary requirements are the initial documentation of the first three chapters. It will commence with the Proposal defense.

Phase III: Capstone Project 2 (August 2026 - December 2026) encompasses the coding of the interactive literacy and jurisprudence modules, asset integration, and reward system logic. This phase concludes with comprehensive system testing, user evaluation in accordance with ISO 25010, and the final oral defense.

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	Phase IV: Post-Defense Revisions and Submission (January 2027 - February 2027) includes incorporating panel feedback from the final defense into the project's documentation and system. Submission of the revised manuscript and securing a final grade from the research course facilitator are key milestones. Phase V: Completion and Clearance (March 2027 - April 2027) encompasses the submission of official receipts for defense and research services and clearance processing. The expected output is the completion of all capstone project requirements.	
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Table 1 – Gantt Chart of SalamLearn

Phase	Activity	2026												2027			
		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr
I	a. Problem Identification & Review of Literature																
	b. Title Proposal & Capsule Submission																
	c. Capsule Acceptance of Research Adviser																
II	a. System Analysis & Requirements Specs																
	b. Writing Chapters 1, 2, and 3																
	c. UI/UX Design (Visual-First, Silent Games) & Database Modeling																
	d. Proposal Defense																
III	a. Back-end Development (Visual Asset Mgmt & Rewards)																
	b. Front-end Integration (Offline modules & Tactile Input)																
	c. System Testing (UI/UX) & ISO 25010 Evaluation																
	d. Final Oral Defense & Documentation																
IV	a. Incorporation of panel feedback.																
	b. Submission of revised manuscript.																
	c. Final grade encoding by facilitator.																
V	a. Submission of official receipts for defense and research services.																
	b. Clearance processing for graduating students.																

1.8 DEFINITION OF TERMS

ALIVE (Arabic Language and Islamic Values Education).

Conceptually, ALIVE is the Department of Education's curriculum framework designed to provide Muslim learners with culturally relevant linguistic and religious education. Operationally, in this study, ALIVE refers to the foundational syllabus and learning objectives that define the content structure of the SalamLearn application.

Gamification. Conceptually, gamification refers to the application of game-design elements, such as points, levels, and badges, in non-game contexts to increase user engagement and motivation. Operationally, in this study, gamification refers to the use of progress badges, rewards, and Halal/Haram drag-and-drop sorting games within the SalamLearn app to incentivize Arabic literacy and religious rule recognition.

Halal / Haram. Conceptually, these are Islamic legal categories defining actions or items as permissible (Halal) or forbidden (Haram) under Sharia law. Operationally, in this study, Halal and Haram refer to the specific binary classifications used in the Rules Gamification Module to teach moral and ethical concepts to children through interactive sorting.

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	Madrasah. Conceptually, a Madrasah refers to a traditional educational institution for Islamic instruction, typically focusing on religious texts, Arabic language, and spiritual values. Operationally, in this study, Madrasah refers to the specific learning environment where SalamLearn is deployed, serving as the primary setting for both individual home-based learning and teacher-facilitated instruction. Offline-First. Conceptually, offline-first is a software architectural strategy where the system remains fully functional and accessible without a continuous or stable internet connection. Operationally, in this study, offline-first refers to the capability of the SalamLearn application to cache all core educational content, audio assets, and progress data locally using Hive, allowing users to access lessons without active cloud connectivity. Tactile Discovery. Conceptually, tactile discovery refers to a pedagogical method where learning is facilitated through physical interaction, touch, and tracing to improve cognitive encoding and motor skill development. Operationally, in this study, tactile discovery refers to the app’s interactive character-tracing module, where students learn to recognize Arabic symbols by tracing them on a touchscreen to receive visual and haptic feedback.	
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	Talqeen. Conceptually, talqeen is a traditional pedagogical method rooted in oral repetition and rote memorization of religious texts and rules. Operationally, in this study, talqeen refers to the existing instructional baseline that SalamLearn intends to supplement rather than replace, providing the digital "bridge" to move from passive listening to active interaction. Telemetry Data. Conceptually, telemetry data refers to the automatic recording and transmission of measurements or performance data from remote sources to IT systems for analysis. Operationally, in this study, telemetry data refers to the collection of student performance metrics, such as tracing accuracy scores and module completion times, which are stored securely to populate the Parent Analytics Dashboard. Whole Class Teaching (WCT). Conceptually, whole class teaching is an instructional strategy where an educator leads group activities or lessons for an entire class of students simultaneously. Operationally, in this study, whole class teaching refers to the "Offline Classroom Package" feature, which enables educators to cast interactive lessons from a single device to projectors or televisions to ensure collective participation in resource-constrained environments.	
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CHAPTER II

REVIEW OF RELATED LITERATURE

2.1 INTRODUCTION TO THE REVIEW

This chapter reviews existing scholarly literature and empirical studies relevant to the design, pedagogical structuring, and technical development of SalamLearn, a gamified mobile learning application designed for foundational Arabic and Islamic education. The rapid integration of digital technology into modern school systems has created many opportunities to improve learning outcomes. However, introducing these technologies within culturally specific and resource-constrained environments requires a highly localized and careful approach. The theoretical foundation of this review connects the Technology Acceptance Model with the Cognitive Theory of Multimedia Learning. This dual-theoretical framework helps explain how mobile learning tools can effectively support traditional religious instruction without disrupting established cultural norms, while ensuring the software remains accessible and easy to use for young learners.

To organize the discussion logically, the literature is grouped into seven distinct thematic areas that provide the technical and pedagogical basis for the SalamLearn system. Theme 1 examines the cognitive profiles and digital literacy milestones of early learners aged 5 to 10, focusing on human-computer interaction and fine motor development on touchscreens. Theme 2 explores Arabic language

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acquisition within the Philippine Department of Education Arabic Language and Islamic Values Education program, illustrating the need to support traditional memorization methods with interactive tools. Theme 3 analyzes the ethical boundaries of gamification and values formation, focusing on the gamification of moral concepts. Theme 4 investigates pedagogical models tailored for resource-constrained classrooms, highlighting Whole Class Teaching and offline-first software architectures. Theme 5 addresses learning analytics, active parental mediation, and compliance with the Philippine Data Privacy Act of 2012 concerning minors' telemetry data. Theme 6 evaluates modern mobile architectures, detailing the advantages of cross-platform frameworks and local databases. Finally, Theme 7 utilizes the ISO/IEC 25010 standard to outline usability and functional suitability metrics for child-centric applications.

Following this thematic review, relevant local and foreign studies are critically analyzed to highlight similarities, methodological differences, and technical limits. The chapter then presents a synthesis that explicitly defines the research gap addressed by this study, before concluding with the theoretical and conceptual frameworks that serve as the blueprint for the proposed system.

2.2 RELATED LITERATURE

Cognitive Profile and Digital Literacy of Early Learners (Ages 5-10)

Understanding how developmental psychology and human-computer interaction connect is extremely important when designing educational apps for young children. Early learners aged 5 to 10 possess distinct cognitive processing

abilities and motor skills that dictate how they interact with touchscreen devices. While traditional classrooms rely heavily on static visual and auditory learning, Hatira (2024) argued that tactile devices engage multiple sensory pathways simultaneously, providing haptic and kinesthetic feedback that improves knowledge retention over static instruction.

Empirical studies on childhood touchscreen gestures show clear age-based developmental milestones that should guide user interface design. Yadav and Chakraborty (2018) assessed touchscreen interactions across various age groups, demonstrating that while toddlers struggle with basic targeting, children aged 4 to 6 successfully perform simple single-finger gestures, such as tapping and sliding, with a 57% success rate. By ages 7 and 8, children graduate to more complex gestures, including dragging and dropping, showing a 30% success rate. Consequently, software designed for early learners should minimize text, avoid complex multi-finger shortcuts, and rely on intuitive, single-finger tactile interactions.

Nacher et al. (2015) verified these limits by showing that writing or drawing on a screen requires significantly more cognitive resources than simple tapping, a capability that begins to develop reliably around age 5. Furthermore, passive screen time can harm motor development. Daud et al. (2020) discovered that intensive, passive touchscreen use correlates with deficits in fine motor precision and manual dexterity compared to non-users. However, these deficits are actively mitigated when the application intentionally requires guided, active tactile interactions.

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This body of literature is highly relevant to SalamLearn because it justifies the design of the interactive character tracing module. By targeting the 5–10 age demographic, the application utilizes single-finger tracing paths and visual cues, which actively builds the child's fine motor coordination while simultaneously reinforcing visual symbol recognition.

Arabic Language Acquisition and Madrasah Curriculum Alignment

Islamic education in the Philippines is deeply rooted in the Madrasah system, which has historically served as the primary institution for preserving Islamic faith, Arabic literacy, and cultural heritage in Muslim-Filipino communities. Sali (2023) explained that these decentralized schools preserve cultural identity, but struggle to connect with the national system. To address this, the Department of Education issued DepEd Order No. 41, s. 2017, establishing guidelines for the Arabic Language and Islamic Values Education program within the K to 12 basic education curriculum. Sapia and Grace (2021) noted that the ALIVE curriculum is designed to support both linguistic proficiency and character development, ensuring public school Muslim learners receive culturally relevant education.

Despite this institutional framework, ALIVE classes face persistent structural and pedagogical challenges. Cortez (2026) explored the lived experiences of Muslim teachers (Asatidz) and found that they frequently encounter difficulties related to inadequate instructional materials, a lack of modern teaching aids, and the challenges of managing multi-grade classrooms. Furthermore, Sali (2023) observed that traditional instruction relies heavily on

talqeen (rote repetition and oral memorization). Mangadang et al. (2025) argued that while talqeen is useful for oral recitation, relying on it exclusively in early childhood causes cognitive fatigue and student disengagement because abstract language rules are taught without interactive or visual reinforcement.

To resolve these limitations, Haron and Asbullah (2025) suggested that integrating localized, curriculum-aligned mobile applications offers a scalable solution to support teachers. Kholifia et al. (2025) researched gamification in Arabic learning and proved that digital rewards and progress tracking reduce language anxiety and boost learner confidence. However, Hussain and Asif (2025) noted that educational apps must follow a structured, curriculum-aligned progression rather than presenting disjointed mini-games to be effective.

These findings directly justify SalamLearn’s structural alignment. By translating the standard DepEd ALIVE curriculum into interactive digital modules, the application functions as a supplementary teaching aid that respects Madrasah traditions while equipping educators with modern, engaging modalities.

Ethical Gamification and Values Formation in Islamic EdTech

Gamification is described as using game-design elements such as points, quest systems, and digital badges in non-game environments. It is shown to be an effective way to boost learner motivation. Wahyuni (2026) studied the impact of gamifying Islamic Religious Education, demonstrating that quest-based structures and digital badges increased active student participation to over 78% and elevated overall comprehension by up to 68%. Safitri et al. (2024) confirmed that digital

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gamification is highly effective for student motivation in religious studies, provided the mechanics do not conflict with religious teachings.

However, applying gamification to faith-based subjects requires strict adherence to ethical and theological boundaries. Zulkefli and Asyraf (2024) emphasized that Islamic EdTech must respect ethical boundaries, avoiding elements linked to gambling (maysir), extreme violence, or prohibited auditory content. Instead, software should use content-neutral mechanics that reward diligence, consistency, and knowledge retention without trivializing sacred material. Sidr Productions (2026) supported using digital rewards as positive reinforcement rather than creating an addictive compulsion.

A critical application of this concept is gamifying moral classifications, such as Halal (permissible) and Haram (forbidden). Syaharani and Muqit (2026) and Tauhid Kids (2026) demonstrated that gamifying moral concepts through visual sorting games and binary logic puzzles allows young children to learn complex ethical rules in a simplified, developmentally appropriate manner. Firdausia (2026) also showed that active digital feedback loops during these sorting and tracing tasks directly support constructivist learning. Mahazari and Hamzah (2026) added that digital tools should feature dual-mode architectures to ensure accessibility in low-infrastructure areas.

This literature directly justifies SalamLearn’s gamified logic. The app avoids addictive features and instead uses ethical gamification, like a Progress Rewards System with digital badges and a Halal/Haram drag-and-drop sorting game to make abstract jurisprudential concepts accessible to early learners.

Pedagogical Models for Resource-Constrained Environments

A major barrier to educational technology in developing, rural, or minority communities is the digital divide. GlobeDock Academy (2023) explained that a severe "engagement gap" occurs in underfunded areas because of a lack of personal devices and reliable internet. Learning Equality (2026) argued that 1-to-1 device ratios are unfeasible in resource-constrained schools, meaning EdTech must leverage shared hardware.

Whole Class Teaching utilizing interactive projection and shared displays provides a highly effective solution to hardware scarcity. As researched by companies like EyeClick (2024) and BenQ (2026), interactive projection systems allow a single device to turn a traditional classroom into a collaborative learning environment, boosting attention span and improving test scores by up to 25%. Karthigesu et al. (2020) pointed out that projecting interactive activities enables shared group participation and bypasses the need for individual smartphones.

To survive in low-connectivity areas, applications must implement offline-first software architectures. Bansod et al. (2026) explained that offline-first principles require software to maintain full functionality without active cloud connections. This is accomplished by caching all critical multimedia assets, audio pronunciations, and curriculum guides directly onto the host device. Learning Equality (2026) and GlobeDock Academy (2023) demonstrated that offline-first educational systems achieve high scalability and sustainability in underserved,

historically marginalized areas. Samudio (2022) noted that developers must balance robust offline data caching with interactive UI designs.

These models form the technical core of SalamLearn. The app features an "Offline Classroom Package" that allows educators to project lessons on a single shared screen, bypassing device scarcity and ensuring uninterrupted learning without an active internet connection.

Learning Analytics, Parental Mediation, and Data Privacy

Deploying mobile applications that collect user telemetry requires a strict approach to data security, especially when dealing with minors. In the Philippines, the Data Privacy Act of 2012 (Republic Act No. 10173) mandates legally binding guidelines for processing personal information. Under Section 3(1) of RA 10173, student academic records, progress logs, and tracking metrics are classified as sensitive personal information. Educational technology developers act as Personal Information Controllers and must enforce data transparency, limit collection, and secure explicit parental consent. Dela Rosa (2025) and Doce (2018) warned that educational software must integrate privacy by design, avoiding third-party advertising trackers and securing local data storage.

Beyond legal compliance, capturing student progress logs feeds into learning analytics, which serves as a powerful diagnostic tool when paired with active parental mediation. Goodall and Flewitt (2025) analyzed digital parenting styles, distinguishing between restrictive mediation (limiting screen access with rigid rules) and active mediation (collaborating with the child to guide learning).

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While restrictive mediation can lead to digital procrastination and limit autonomy, active mediation improves cognitive outcomes. Ma and Li (2026) verified that rigid, restrictive locks fail to correct mobile usage habits, making active guidance the preferred strategy.

To support active mediation, mobile apps must provide parental analytics dashboards. These dashboards translate raw telemetry, such as tracing accuracy, module completion times, and jurisprudence sorting scores into visual charts. This data-driven transparency allows parents to monitor learning progress asynchronously. Armed with these logs, parents can offer targeted praise and identify unique learning difficulties, actively participating in their child's education without compromising the child's data privacy rights.

This justifies the design of SalamLearn’s data systems. The app complies with RA 10173 by localizing student progress data and securing parental consent, while its Parent Analytics Dashboard equips caregivers with visual progress logs to support active home mediation.

Evaluating Mobile Architectures in EdTech

The technical architecture of a mobile application directly determines its performance, frame rates, and longevity on older devices. For community-oriented projects targeting varied hardware, cross-platform mobile frameworks are essential. Panda (2025) compared cross-platform options and highlighted Google's Flutter framework as a dominant tool because of its single-codebase development model using Dart.

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Unlike older cross-platform frameworks like React Native, which communicate asynchronously with native hardware components via a JavaScript bridge. Flutter compiles Dart code Ahead-of-Time directly into native ARM machine code. Foresight Mobile (2026) documented that bypassing the JavaScript bridge eliminates rendering lag, allowing Flutter to render user interfaces at 60 to 120 FPS with cold start times under 200 ms. For child-centric apps requiring real-time tracing modules and physics-based sorting animations, Flutter's high-performance rendering engine (Skia/Impeller) is critical to prevent visual stuttering and maintain user immersion.

Equally critical is selecting an offline database. Ibrahim Aboelyazed (2025) contrasted SQLite, a traditional relational database, with Hive, a lightweight NoSQL key-value store built specifically for Dart. Relational databases are highly effective for complex, multi-table queries, but Hive is optimized for rapid read/write operations and has no native dependencies. Ochoa and España (2026) and Zohud and Zein (2021) validated that the Flutter and Firebase stack provides high usability and reduces development cycles.

This stack directly supports SalamLearn’s offline-first performance. By building the front-end with Flutter, the app renders fluid character tracing animations, while its hybrid database architecture uses Hive for instant local caching and Firebase for secure parental sync.

Software Quality Standards (ISO/IEC 25010)

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To ensure that an educational application meets its pedagogical goals and runs reliably on target devices, developers must evaluate it against established international standards. The ISO/IEC 25010 standard defines a quality framework composed of eight characteristics, with Usability and Functional Suitability being paramount for child-centric software.

Functional suitability evaluates whether the application provides accurate, complete, and appropriate features for its goals. For SalamLearn, this ensures that Arabic characters are drawn, traced, and pronounced accurately according to curriculum guidelines, and that gamified rules correctly teach Halal and Haram concepts.

Usability (revised as interaction capability in the 2023 standard update) measures how effectively and satisfyingly a user can operate the software. Jung et al. (2025) analyzed early childhood app feedback and mapped user concerns directly to ISO/IEC 25010. Their analysis showed that usability, functional suitability, and compatibility represent parents' primary concerns regarding educational reliability and ease of use. Evaluating usability with children aged 5 to 10 requires objective testing methods. Carrión-León et al. (2025) proved that traditional self-reported surveys fail when used with young children. Instead, researchers must capture objective gameplay telemetry, capturing touch coordinates, time-on-task, and error recovery rates to accurately measure usability. Finally, Nugroho and Fx (2025) noted that software architectures must prioritize maintainability, ensuring that codebases are structured for easy modifications when adding advanced Madrasah modules.

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This guides SalamLearn's evaluation methodology. The app will be assessed by IT experts and educators using ISO/IEC 25010 usability metrics, and user testing with children will rely on objective gameplay telemetry rather than complex surveys.

2.3 RELATED STUDIES

The review of related studies contextualizes the proposed project within the existing body of empirical research, identifying how previous developers, educators, and academicians have tackled similar educational and technological challenges both locally and internationally. This comparative analysis serves to highlight established best practices while simultaneously exposing the functional limitations that the present study intends to resolve.

Local Studies

In the Philippines, several completed studies have explored the practical and pedagogical aspects of the DepEd ALIVE program, as well as the implementation of offline educational technology in rural settings. Mangadang et al. (2025) conducted a descriptive study to assess the implementation of the ALIVE program in public elementary schools across Mindanao. Their findings revealed that public schools struggle with a severe lack of localized visual media and short instructional periods, which limits Arabic vocabulary retention among young learners. This study is highly relevant because both projects align with the ALIVE curriculum to improve Arabic literacy, but Mangadang et al. (2025)

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focused on program assessment, whereas SalamLearn serves as a practical software intervention.

Cortez (2026) conducted a qualitative phenomenological study exploring the lived experiences of Asatidz teaching ALIVE in public schools. The study found that teachers face difficulties due to multi-grade classrooms, inadequate instructional aids, and a lack of technological support. This qualitative research aligns with the current study's focus on the public school classroom environment, but Cortez (2026) focused on documenting teaching experiences, while SalamLearn provides an "Offline Classroom Package" to directly support these educators.

Addressing technical limitations in rural schools, Bansod et al. (2026) developed and evaluated an offline-first examination and data management system for rural school societies. Utilizing an agile development methodology, the researchers demonstrated that local database persistence and reactive frontend caching maintain high operational efficiency in low-connectivity areas. Both systems use offline-first databases to bypass poor internet connectivity, but Bansod et al. (2026) focused on administrative examination data, whereas SalamLearn targets early childhood gamification and character tracing.

Dela Rosa (2025) and Doce (2018) conducted quantitative evaluations of state university compliance with the Data Privacy Act of 2012 (RA 10173). Their audits showed that educational information systems frequently compromise student data because of technical constraints and a lack of privacy-by-design training. While these studies emphasize the critical need for data protection, they

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	analyzed administrative university portals, whereas SalamLearn is a mobile application designed specifically to secure the telemetry data of minors. Mina (2021) evaluated a local online enrollment system using the ISO/IEC 25010 quality framework. Using survey evaluations, the research showed that while the system scored high in functional completeness, it struggled with performance efficiency because of local bandwidth bottlenecks. Both studies evaluate educational software using international ISO quality standards, but Mina (2021) evaluated an online enrollment platform, while SalamLearn relies on local Hive caching to bypass bandwidth limits entirely. Finally, Abrenica (2021) conducted a qualitative case study on the academic challenges faced by Muslim learners in public high schools. The research showed that minority students face academic anxiety and strongly desire culturally specific curriculum integration to support their religious identity. Both projects recognize the need to support Muslim student identities, but Abrenica (2021) analyzed qualitative high school student perspectives, whereas SalamLearn is a mobile tool designed for foundational childhood literacy. Foreign Studies Internationally, the digitization and gamification of Arabic language acquisition and Islamic studies have progressed rapidly, providing functional blueprints. Al Hejaili and Newbury (2023) developed and evaluated LAA: Learn the Arabic Alphabet, a touchscreen mobile application. Using user experience testing with young cohorts, they demonstrated that gamified on-screen character	
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tracing and visual-auditory matching improve letter recognition. This is similar since it serves as the direct inspiration for SalamLearn’s tactile tracing module. However, their app was a single-learner tool, while SalamLearn includes an offline classroom casting mode and broader Islamic jurisprudence games.

Kholifia et al. (2025) evaluated gamified Arabic speaking tools for language anxiety. Using a quasi-experimental method, they discovered that interactive progress rewards and digital badges lower student anxiety and improve engagement. Both use gamified mechanics to encourage language acquisition. However, Kholifia et al. (2025) focused on oral speaking skills for older learners, while SalamLearn targets basic writing and spelling for children aged 5–10.

Wahyuni (2026) investigated gamified Islamic Religious Education (PAI) in Indonesia. Using a quasi-experimental research design, the study showed that quest structures and points raised student interest and improved comprehension by 68. Both projects gamify religious values. However, Wahyuni (2026) conducted her study in an Indonesian curriculum context requiring 1-to-1 personal tablets, whereas SalamLearn is built for the Philippine ALIVE framework and supports a 1-to-many classroom projection model.

2.4 SYNTHESIS AND RESEARCH GAP

The reviewed literature and related studies consistently highlight the benefits of gamification, tactile human-computer interaction, and mobile learning in early childhood education. Foreign studies have shown that gamifying Arabic character-tracing and Islamic jurisprudence reduces cognitive load, lowers

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	language anxiety, and improves long-term memory. At the same time, local studies highlight the persistent challenges in the Philippine Madrasah and ALIVE systems, where teachers struggle with a lack of resources, multi-grade classrooms, and poor internet access. However, existing foreign educational applications are built for Middle Eastern or Indonesian classrooms and require 1-to-1 student device access with continuous internet connectivity, making them unusable in underfunded public schools. Conversely, local software solutions focus primarily on administrative features and lack interactive, culturally appropriate gamified content designed specifically for Filipino Muslim children. This exposes a specific research gap: there is currently no localized, high-performance, cross-platform mobile application designed specifically for the Filipino Muslim child (ages 5–10) that integrates tactile Arabic tracing and Halal/Haram gamification while simultaneously functioning as an offline-first, Whole Class Teaching (WCT) casting tool for resource-constrained Madrasahs, all while maintaining strict compliance with the Philippine Data Privacy Act (RA 10173) for the telemetry data of minors. SalamLearn directly addresses this gap. It utilizes Flutter and Hive to build an offline-first mobile app that features a progress dashboard for active parental mediation and an offline projection mode for teachers, permanently bridging the digital divide in Filipino Madrasah education. 2.5 THEORETICAL FRAMEWORK	
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This study is anchored on two primary theories: the Technology Acceptance Model (TAM) proposed by Fred Davis (1989), and the Cognitive Theory of Multimedia Learning developed by Richard Mayer (2001). Together, these theories explain why users will adopt the proposed software and how children will process its educational content.

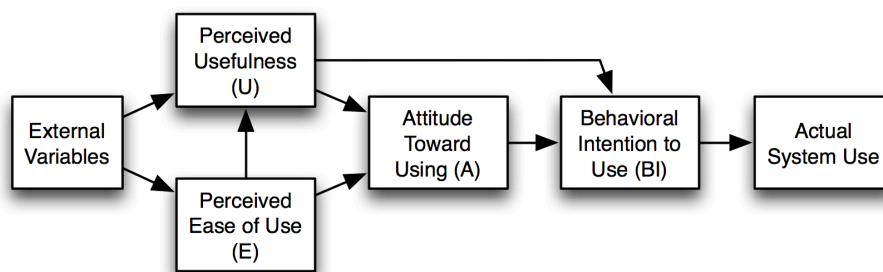


Figure 1. Technology Acceptance Model

The Technology Acceptance Model states that technology adoption is driven by Perceived Usefulness (PU) and Perceived Ease of Use (PEOU). For SalamLearn, Perceived Usefulness directly affects teachers and parents. For Asatidz, usefulness is established because the app provides an Offline Classroom Package, allowing teachers to cast interactive group lessons and bypass hardware scarcity. For parents, usefulness is demonstrated by the analytics dashboard, which allows them to track progress and practice active mediation. Perceived Ease of Use centers on the application's interface design. To be accepted, the child's interface must use single-finger gestures, avoid complex multi-finger

shortcuts, and utilize intuitive visual-first navigation to accommodate early learners with developing cognitive abilities.

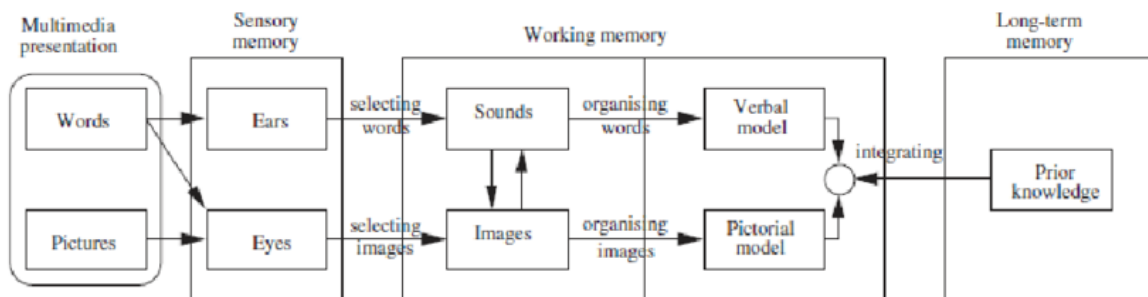


Figure 2. Mayer's Cognitive Theory of Multimedia Learning

Mayer's Cognitive Theory of Multimedia Learning supports TAM by explaining how young children process digital information. The theory states that the human brain processes visual and auditory information through separate cognitive channels, and that learning is optimized when both channels are stimulated simultaneously without overloading working memory. SalamLearn applies this theory through its multi-sensory modules: combining interactive path-tracing (kinesthetic), colorful character graphics (visual), and authentic audio pronunciations (auditory). Designing the app around these dual channels prevents cognitive fatigue while maximizing retention, ensuring that the interface is developmentally appropriate and pedagogically sound under ISO/IEC 25010 standards.

2.6 CONCEPTUAL FRAMEWORK

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The conceptual framework of this study is structured around the Input-Process-Output (IPO) model. This framework maps the inputs, agile processes, and system outputs required to build a validated educational technology solution for Filipino Madrasah environments.

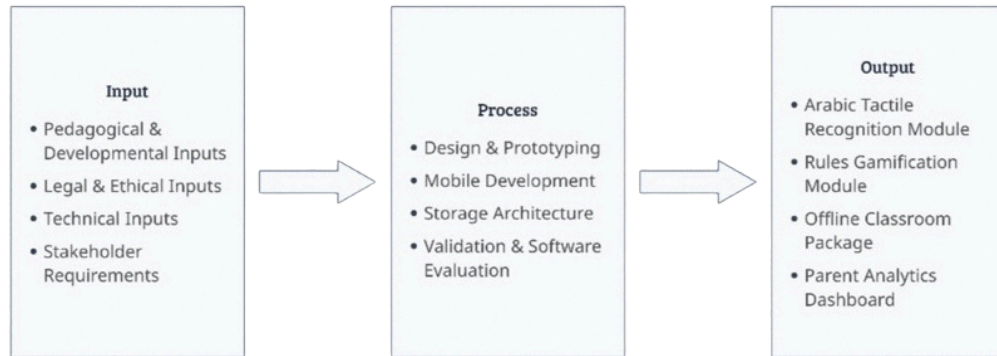


Figure 3. Input-Process-Output Model of SalamLearn

Input

The inputs represent the pedagogical, developmental, legal, and technical requirements needed to start development :

- **Pedagogical & Developmental Inputs:** The Refined Elementary Madrasah Curriculum guidelines and DepEd ALIVE standards form the core syllabus. These are paired with child HCI guidelines mapping touchscreen motor skills for children aged 5 to 10 to ensure a low cognitive load.
- **Legal & Ethical Inputs:** Legal requirements from the Philippine Data Privacy Act of 2012 (RA 10173) dictate how the app must collect and store minors' sensitive performance data.

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- **Technical Inputs:** The cross-platform Flutter framework and local key-value storage databases (Hive) provide the development foundation.
- **Stakeholder Requirements:** Initial user feedback gathered from public school Asatidz and Muslim parents identifies structural barriers, like a lack of Wi-Fi and school device bans.

Process

The process phase translates these inputs into a functional solution using an iterative software development lifecycle :

- **Design & Prototyping:** Creating wireframes that prioritize large buttons, high-contrast colors, and visual animations, completely avoiding text-heavy menus.
- **Mobile Development:** Developing the application with Google's Flutter framework to ensure 60 to 120 FPS performance, compiled Ahead-of-Time directly into native Android code to eliminate JavaScript rendering lag.
- **Storage Architecture:** Writing local caching scripts in Hive to store all character animations and audio clips, ensuring full offline functionality. This is paired with secure Firebase synchronization scripts that update the parental dashboard when the device connects to the internet.
- **Validation & Software Evaluation:** Submitting the Arabic pronunciations, daily guides, and Halal/Haram classifications to local Asatidz for content validation. This phase culminates in usability testing with child cohorts, measuring interaction metrics, such as touch accuracy

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and time-on-task to evaluate usability, compatibility, and functional suitability under ISO/IEC 25010 standards.

Output

The final output is a validated, deployed version of the SalamLearn mobile application, tailored for the Filipino Madrasah context :

- **Arabic Tactile Recognition Module:** An interactive interface where children learn to draw Arabic letters using single-finger path-tracing games.
- **Rules Gamification Module:** A drag-and-drop sorting game designed to teach the classifications of Halal and Haram.
- **Offline Classroom Package:** A toolkit that allows Asatidz to cast interactive lessons using projectors or shared displays, bypassing the need for individual student smartphones.
- **Parent Analytics Dashboard:** A secure, RA 10173-compliant progress monitor that translates raw gameplay telemetry into visually intuitive charts. This dashboard helps parents practice active, asynchronous mediation to reinforce learning at home.

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